

NAME (PRINT):

Last/Surname

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STUDENT #:

SIGNATURE:

**UNIVERSITY OF TORONTO MISSISSAUGA
APRIL 2019 FINAL EXAMINATION
CSC343H5S
Introduction to Databases
Michael Liut
Duration - 3 hours
Aids: None**

The University of Toronto Mississauga and you, as a student, share a commitment to academic integrity. You are reminded that you may be charged with an academic offence for possessing any unauthorized aids during the writing of an exam. Clear, sealable, plastic bags have been provided for all electronic devices with storage, including but not limited to: cell phones, smart watches, SMART devices, tablets, laptops, and calculators. Please turn off all devices, seal them in the bag provided, and place the bag under your desk for the duration of the examination. You will not be able to touch the bag or its contents until the exam is over.

If, during an exam, any of these items are found on your person or in the area of your desk other than in the clear, sealable, plastic bag, you may be charged with an academic offence. A typical penalty for an academic offence may cause you to fail the course.

*Please note, once this exam has begun, you **CANNOT** re-write it.*

To pass this course you are required to obtain at least 40% on this examination.

Additional instructions are located on page 3 of this exam booklet. Please read them prior to starting the exam.

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NAME: _____ ID: _____

CSC 343 H5S

Day Class 01

Instructor: Michael Liut

DURATION: 3 hours

University of Toronto Mississauga

Final Examination

April 2019

Read all instructions before starting the exam

This test paper includes 26 pages and a total of 27 questions; each question requires a written answer, even justification for the true-or-false. You are responsible for ensuring that your copy of the paper is complete. Bring any discrepancy to the attention of your invigilator.

Special Instructions:

1. The answers to all questions must be written in the space provided (i.e. immediately following the question). Written answers are to be clear and concise; illegible solutions will not be marked and awarded a grade of 0.
2. Read all questions before starting the exam - some are easier than others.
3. **This is a closed book exam.** No memory aids or textbooks of any kind are allowed during the test. Supporting pages have been appended to this exam.
4. The use of any electronic device is strictly prohibited. Failure to comply will result in your immediate dismissal from the examination and a grade of 0.
5. The students are not allowed to be involved in any communication of any kind concerning the questions and answers of this exam with anybody except the invigilator(s) or the instructor. Any attempt of such communication will be considered a case of academic dishonesty.
6. No unauthorized scrap paper is allowed to be used. The invigilator(s) will supply every student with needed scrap paper when asked.
7. Documents to be returned: this questionnaire (i.e. the exam) and, if used, all scrap paper. All documents must bear the student's name and student's identification number. Only the face page of the questionnaire needs to bear the student's name and number.
8. On this page, to the right of this line, draw an animal of your choice and give it a name. By doing so you will be awarded with one additional point on this exam.

question	mark	out of
1–10		11
11–14		7
15–21		7
22		4
23		16
24		8
25		6
26		10
27		6
total		75

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Questions 1 – 10 are True-or-False questions. You are required to illustrate the validity by an example, or disprove the validity by a counter-example

For each question only select one answer (i.e. True or False). To select an answer, you must circle the word “True” or the word “False”. Incorrect or missing answers earn a mark of 0. Therefore, do not leave any questions blank. For full marks you must justify (prove/disprove) your answer.

I. True or False (11 marks)

Question 1 [2 marks] Consider the same relational schema R given in Assignment 3. R has a set of attributes $\alpha = \{A, B, C, D, E, F, G, H\}$ and the set of functional dependencies $\mathcal{F} = \{A \rightarrow B, ABD \rightarrow FGH, AEH \rightarrow BD, BC \rightarrow EH, C \rightarrow ACG, C \rightarrow AFH, DE \rightarrow HB, DF \rightarrow AC, E \rightarrow F, H \rightarrow EA\}$.

Through closure, we are able to obtain the following Candidate Keys: $\{\{C\}, \{H\}, \{A, D\}, \{D, E\}, \{D, F\}\}$.

- i. [1 mark] Given R , with α and $\mathcal{F}$, a 3NF decomposition would result in the same candidate keys found by computing the closure of each attribute in α .

- A. True.
B. False.

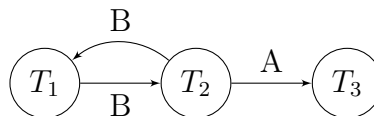
- ii. [1 mark] Given R , with α and $\mathcal{F}$, there can only be one unique 3NF decomposition.

- A. True.
B. False.

Question 2 [1 mark] The following schedule:

$R_2(A); R_1(B); W_2(A); R_2(B); R_3(A); W_1(B); W_3(A); W_2(B);$

has the following precedence graph:



thus, making this schedule conflict serializable.

- A. True.
B. False.

Question 3 [1 mark] Given a relation $R(A,B,C,D)$, with this data set¹:

R:

A	B	C	D
1	3	9	5
2	3	9	5
1	3	4	5
5	7	9	5

prove if the following statements are true or disprove if they are false:

- i. The relation has the following functional dependencies: $CD \rightarrow B$ and $C \rightarrow A$.
 - A. True.
 - B. False.

- ii. Based on R 's instance of data, it is possible that in general you can identify functional dependencies for this relation.
 - A. True.
 - B. False.

Question 4 [1 mark] Let's assume that we are working with subset of a banking schema. Given two of the entities: **Customer** and **Loan**, and their relationship **Borrower**, it is fair to assume that there is a partial participation between **Customer** **Borrower** and clear to assume a total participation between **Borrower** — **Loan**.²

- A. True.
- B. False.

Question 5 [1 mark] Fine-grained locking results in a lower degree of concurrency and an increase in locking overhead versus coarse-grained locking.

- A. True.
- B. False.

¹we assume that this is the entirety of the set, do not consider the 'general case'

²real definition: indicates partial participation and — indicates total participation

Question 6 [1 mark] Virtual views are constructed and stored. Further, virtual views may be periodically updated depending on their query definition.

- A. True.
- B. False.

Question 7 [1 mark] Let R represent a relational schema with a set of attributes α . Let α_1 and α_2 represent two subsets of α , where we say $\alpha_1 \rightarrow \alpha_2$ (assuming the values of α_2 are determined by α_1). More precisely, for any two tuples t_1 and t_2 , in any instance of R , we can say:

$$\Pi_{\alpha_1}(t_1) = \Pi_{\alpha_2}(t_2) \implies \Pi_{\alpha_1}(t_1) = \Pi_{\alpha_2}(t_2)$$

- A. True.
- B. False.

Question 8 [1 mark] Given a set of Functional Dependencies $\mathcal{F}$, the term “Dependency Preservation” is formally defined as $\mathcal{F}^+ = (\mathcal{F}_1 \cup \dots \cup \mathcal{F}_n)^+$, where $\mathcal{F}_1 \dots \mathcal{F}_n$ represent schemas with attribute sets of $1..n$.

- A. True.
- B. False.

Question 9 [1 mark] Clustered indexes are more expensive to maintain than a fully sorted file.

- A. True.
- B. False.

Question 10 [1 mark] Strict Two-Phase Locking (S2PL) and non-strict Two-Phase Locking (2PL) both ensure acyclicity of the precedence graph and, therefore, allow only conflict serializable schedules.

- A. True.
- B. False.

Questions 11 – 14 are multiple-choice questions.

For each question only circle one answer. Incorrect or missing answers earn a mark of 0. Therefore, do not leave any questions blank.

II. Multiple Choice (7 marks)

Question 11 [4 marks] If a Transactional Process System (TPS)³ is demonstrating the following behaviours, which transactional property is said to be violated in each of the following situations:

- i. the disk crashes in a thunderstorm and all the data stored in the database is lost.
 - A. Atomicity.
 - B. Isolation.
 - C. Consistency.
 - D. Durability.
 - E. None of the properties are violated.

- ii. two concurrently executing transactions write to the same data item.
 - A. Atomicity.
 - B. Isolation.
 - C. Consistency.
 - D. Durability.
 - E. None of the properties are violated.

- iii. the system crashes while executing a script of transactions; the partial changes made by the script are not rolled back.
 - A. Atomicity.
 - B. Isolation.
 - C. Consistency.
 - D. Durability.
 - E. None of the properties are violated.

- iv. the state of the database, after executing a transaction, is different from the state of the real world.
 - A. Atomicity.
 - B. Isolation.
 - C. Consistency.
 - D. Durability.
 - E. None of the properties are violated.

³**TPS:** a is a type of information system that collects, stores, modifies and retrieves the data transactions of an enterprise.

Question 12 [1 mark] Consider a relation **Supplier(sno, status, city)**. The only candidate key is **sno** and the functional dependency **city → status** exists. Ignoring 1NF and 2NF, what normal form does the relation satisfy?

- A. The relation **Supplier** has no superkey, violating entity-relationship integrity.
- B. The relation **Supplier** is not in 3NF.
- C. The relation **Supplier** is in 3NF, but not in BCNF.
- D. The relation **Supplier** is in BCNF.
- E. The question does not make sense, there is not enough information to decide.

Question 13 [1 mark] Consider a relation **Address(aid, streetNumber, streetName, city, postalCode, province)**, where PRIMARY KEY(**aid**) and UNIQUE(**streetNumber, streetName, city**). In addition to all the obvious keys, we are told that **postalCode → streetName**, **postalCode → city**, and **postalCode → province**. Given this information which of the following statements is most correct?

- A. The relation **Address** is in 3NF.
- B. The relation **Address** is not in 3NF, because **postalCode → streetName** violates 3NF.
- C. The relation **Address** is not in 3NF, because **postalCode → city** violates 3NF.
- D. The relation **Address** is not in 3NF, because **postalCode → province** violates 3NF.
- E. The question does not make sense, there is not enough information to decide.

Question 14 [1 mark] Consider the relation **Address**, defined above in Question 13. What does the SQL statement **UNIQUE(streetNumber, streetName, city)** best express?

- A. It designates the set of attributes {**streetNumber, streetName, city**} as a (secondary) key of the relation **Address**.
- B. It is a constraint requiring that each value of any of the three attributes listed to be unique in the table; e.g. there can be at most one row with the value **streetNumber = 99** or can be at most one row with the value **streetName = 'Main St.'**.
- C. It is a constraint requiring uniqueness of the triple of values for {**streetNumber, streetName, city**}.
- D. Two of the above.
- E. All of the above.

Question 15 – 21 are fill-in-the-blank questions.

For each blank only provide one answer. Incorrect or missing answers earn a mark of 0. Therefore, do not leave any questions blank. A word bank has been appended.

III. Fill in the Blank (7 marks)

Question 15 [1 mark]

Two types of intra-relational constraints are: _____ constraints and _____ constraints.

Question 16 [1 mark]

If all foreign key constraints are enforced then _____ is achieved.

Question 17 [1 mark]

Given two relations R and S , when joined there exists a tuple of R which has no natural match in S . That tuple is said to be _____. Further, that tuple may be joined, but will be _____ by NULL.

Question 18 [1 mark]

_____ is a term used to describe one of the fundamental uses of DBMSs. It also allows us to treat a relationship set as an entity set for the purposes of participation in (other) relationships.

Question 19 [1 mark]

A _____ key is a minimal key that allows us to uniquely identify all tuples in a relation. The database administrator/architect selects one of those keys, as there can be many, and makes a decision on which will be the _____ key by arbitrary or strategic selection.

Question 20 [1 mark]

Given two relations R and S ; R has n rows and c columns, and S has m rows and k columns. On output this means that $R \times S$ will produce $n \times m$ _____ and $c + k$ _____.

Question 21 [1 mark]

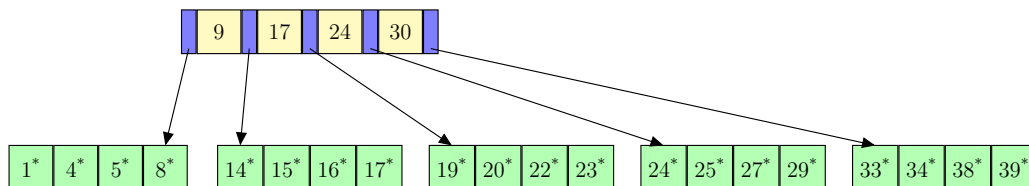
A _____ is easy to understand as they are not operationally defined and prevent inconsistency by any kind of SQL statement while a _____ alerts the user and/or DBA to unusual events and can even create a log of events to support both auditing and security checks.

Questions 22 – 27 are questions that require a written answer. The answers are to be written in the space provided below each question.

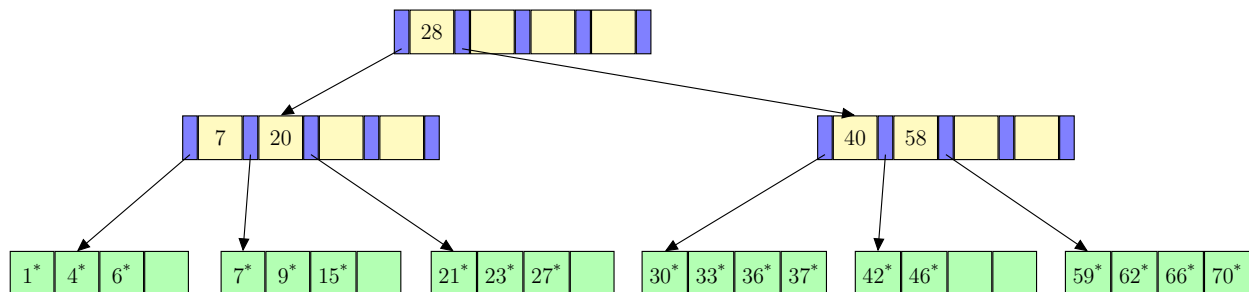
IV. Short Answer (50 marks)

Question 22 [4 marks] Given the following B+ Trees, complete the following action:

- a. Insert 21 into the B+ Tree below.



- b. Delete 46 from the B+ Tree below.



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Question 23 [16 marks] Given the following schema:

Flights (flno:integer, from:string, to:string, distance:integer,
 departs:time, arrives:time, price:integer)
Aircraft (aid:integer, aname:string, cruising_range:integer)
Certified (eid:integer, aid:integer)
Employees (eid:integer, ename:string, salary:integer)
Assigned (flno:integer, eid:integer, last_assigned:datetime)

1. [6 marks] Draw an ERD (using Chen's notation) that represents the schema above.
2. [5 marks] Write an SQL query that finds the names of pilots who can operate planes with a cruising range greater than 3,000 kilometres but are not certified on any "Boeing" aircraft.
3. [5 marks] Translate your SQL query to Relational Algebra.

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Question 24 [8 marks] Consider a relation R with a set of attributes $\alpha = \{A, B, C, D, E, F\}$ and the functional dependency $\mathcal{F}_1 = \{A \rightarrow BCDEF\}$, that holds over R . Define two other functional dependencies $\mathcal{F}_2$ and $\mathcal{F}_3$, which satisfy the following three properties:

- i. Neither $\mathcal{F}_2$ nor $\mathcal{F}_3$ can be inferred from $\mathcal{F}_1$ using Armstrong's Axioms.
- ii. Relation R , with functional dependencies $\mathcal{F}_1$ and $\mathcal{F}_2$, is in BCNF.
- iii. Relation R , with functional dependencies $\mathcal{F}_1$, $\mathcal{F}_2$, and $\mathcal{F}_3$, is in 3NF but not BCNF.

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Question 25 [6 marks] For each of the following locking protocols (a–d below), assuming that every transaction follows that specific locking protocol, explicitly state which of these desirable properties are ensured: serializability, conflict-serializability, recoverability, avoidance of cascading aborts.

- a. Always obtain an exclusive lock before writing; hold exclusive locks until the end-of-transaction. No shared locks are ever obtained.
- b. In addition to protocol a, obtain a shared lock before reading; shared locks can be released at any time.
- c. In addition to protocol b, which encompasses protocol a, locking is two-phase.
- d. In addition to b, which encompasses protocol a, all locks are held until the end-of-transaction.

Answers are to be selected in the table below; you are to use a check mark “✓” to identify which desirable properties are guaranteed and an “X” to identify the ones which are not guaranteed. For those that cannot be determined mark them with a “?” and add an explanation.

Property Question	Serializable	Conflict-Serializable	Recoverable	Avoids Cascading Aborts
a.				
b.				
c.				
d.				

Question 26 [10 marks] Consider the following sequences of actions, listed in the order in which they are submitted to the DBMS:

Sequence S1: $T_1:R(A)$, $T_2:W(A)$, $T_2:W(B)$, $T_3:W(B)$, $T_1:W(B)$,
 $T_1:Commit$, $T_2:Commit$, $T_3:Commit$

Sequence S2: $T_1:R(A)$, $T_2:W(B)$, $T_2:W(A)$, $T_3:W(B)$, $T_1:W(B)$,
 $T_1:Commit$, $T_2:Commit$, $T_3:Commit$

For each sequence and for each of the following concurrency control mechanisms, describe (in detail) how the concurrency control mechanism handles each sequence.

Assume that the timestamp of transaction T_i is i . For lock-based concurrency control mechanism, add lock and unlock requests to the above sequence of actions as per the locking protocol. If a transaction is blocked, assume that all of its actions are queued until it is resumed; the DBMS continues with the next action (according to the listed sequence) of an unblocked transaction.

1. Strict 2PL with timestamps used for deadlock prevention.

Note: implement the “Wait-Die” policy.

- i. How does the concurrency control mechanism handle **Sequence S1**?
- ii. How does the concurrency control mechanism handle **Sequence S2**?

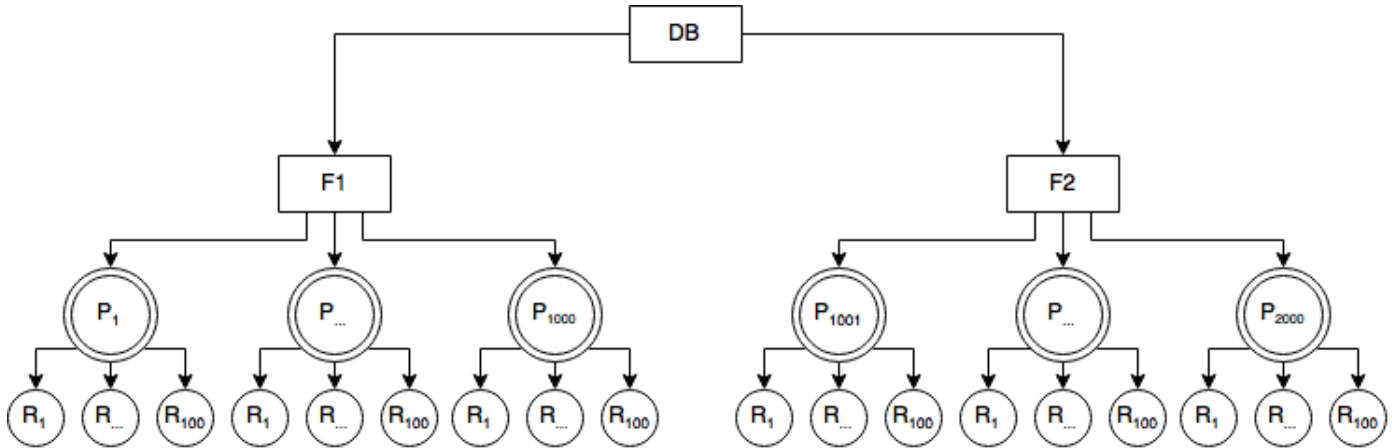
2. Strict 2PL with deadlock detection. Show the waits-for graph if a deadlock cycle develops.

- i. How does the concurrency control mechanism handle **Sequence S1**?
- ii. How does the concurrency control mechanism handle **Sequence S2**?

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Question 27 [6 marks] Consider a database organized in the following hierarchy of objects: The database is an object DB, and it contains two files F1 and F2, each of which contain 1000 pages P. Every page has 100 records R. This is illustrated below, where $P_{...}$ and $R_{...}$ represents all the pages and records between the left most node in the file/page and the right most node in the file/page.



With the use of this hierarchy, *Multiple-Granularity Locking* is used, with S, X, IS, IX and SIX locks, and the database-level, file-level, page-level, and record-level locking.

1. Read P_{10} through P_{980} .
 - (a) What are the sequence of lock requests that must be generated by a transaction that wants to carry out these operations?
 - (b) What can you tell me in terms of performance versus transaction blocking?
2. Delete the first Record R from each page. (HINT: These are all blind writes.)
 - (a) What are the sequence of lock requests that must be generated by a transaction that wants to carry out these operations?
 - (b) What is the trade-off we face with performing this transaction?

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Fill-in-the-Blank Word Bank

Abort
Access
Accessible
Acyclic
Aggregation
Arity
Applied
Assertion
Atomicity
Automatic
Automation
Between
Cache
Candidate
Cardinality
Cascade
Check
Classification
Columns
Commit
Consistency
Constraint
Cross Product
Cursor
Dangling
Daemon
Degree
Denormalization
Difference
Distinct
Distribution
Division
Domain
Dormant
Durability
E-R Integrity
Fragmentation
Git
Graph
Index
Industrial
Intersection
I-R Integrity

Isolation
Join
Kernel
Key
Lock
Log
Magic
Manipulation
Manual
Mastery
Method
Monopoly
Mutual exclusion
Normalization
Null
OLAP
Optimization
Padded
Performance Analysis
Phase of recovery
Precedence graph
Primary
Prime
Process
Production
Programming
Projection
Range
Read
Record
Referenced Integrity
Referential Integrity
Regression
Rollback
Rollup
Rows
Safety
Secondary
Selection
Sequential
Sequestered
Session
SOAP

Software
Sorted files
Sorting
Sound Axiom
Split operator
Stack overflow
Standardization
Static Indexes
Stored Procedures
Stratification
Structured
Substitution
Super
System
Telepathic
Tertiary
Theta-Join
Throughput
Trail
Trigger
Tuple
Union
Unique
Unknown
Vertical fragmentation
Wait-die
Waits-for graph
Workflow management
Wound-wait
Write

REFERENCE SHEET

Chen's Notation				
		Participations Cardinality can be shown or hidden	Recursive Relationship Cardinality can be shown or hidden	
Entity	Attribute	Mandatory		
Weak Entity	Key attribute			
Relationship	Weak key attribute			
Identifying Relationship	Derived attribute	Optional		
Associative Entity	Multivalue attribute			

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